Orthogonal Finetuning for Chinese Instruction

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Project repository: `https://github.com/SleepyLGod/mini-oft-llm.git`

1 Introduction

This report studies Orthogonal Finetuning (OFT) for Chinese instruction-following supervised finetuning on `Qwen2.5-7B-Instruct`. The assignment requires training-loss curves and final quantitative and/or before-vs-after qualitative outcomes. The report is designed to satisfy those requirements with evidence-driven claims only.

Background and motivation. OFT applies multiplicative orthogonal updates, offering a parameter-efficient alternative to additive low-rank adaptation such as LoRA [Qiu et al., 2023, Hu et al., 2021]. The motivation is to improve instruction behavior under fixed compute and limited training window, while avoiding full-parameter finetuning cost. In practical terms, this work asks whether a constrained OFT update can produce robust in-domain gains without expanding adaptation complexity.

Experiment objectives. Effectiveness-wise, we want to measure in-domain likelihood improvement of OFT over the frozen base model. Sensitivity-wise, we want to compare `oft_block_size=32` against `16` under matched setup.

The report evaluates both objectives using quantitative and qualitative evidence jointly: token-level likelihood for global trend, and fixed prompts for behavioral strengths/failures. This combination is important because single-number improvement does not guarantee uniform instruction-following quality.

2 Experimental Design and Implementation

Data. A Chinese instruction corpus is used with fixed split and seed. Final split sizes are 1,484,455 (train), 82,465 (validation), and 82,479 (test), totaling 1,649,399 samples [YeungNLP, 2023].

Model and compute. The base model is Qwen2.5-7B-Instruct [Qwen Team, 2024]. Training is distributed over 2×H100 NVL in bf16.

Training setup. Main run and ablation run both use 1200 steps with matched optimizer/scheduler/batch settings. OFT is applied to all linear modules with Cayley-Neumann enabled. The only ablated variable is `oft_block_size` (32 vs. 16).

Instruction tuning is implemented as causal language-model SFT with unified sequence formatting, so all compared runs optimize the same objective on the same tokenization regime. This removes task-definition confounders from the ablation comparison.

Implementation pipeline details. The workflow is fixed to reduce execution risk: deterministic preprocessing, a 100-step distributed smoke run, two matched full runs (main and ablation), and a shared evaluator on the same 2000-sample test slice. Quantitative metrics and before/after generations are then consolidated for consistency checks between likelihood gains and behavior changes.

Evaluation protocol. Three evidence channels are used:

Table 1: Primary quantitative results on 2000 held-out samples (lower is better). The table supports in-domain likelihood comparison, not out-of-domain generalization claims.

Model	Mean NLL	PPL	ΔNLL vs Base	ΔPPL vs Base	eval_loss
Base (no finetuning)	3.2619	26.0990	–	–	–
OFT main (`block_size=32`)	**1.9770**	**7.2207**	**-39.39%**	**-72.33%**	**1.7879**
OFT ablation (`block_size=16`)	1.9988	7.3803	-38.72%	-71.72%	1.8102

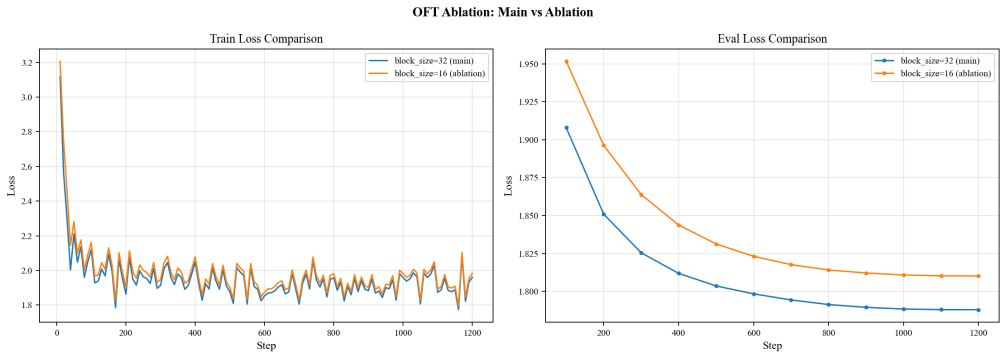

Figure 1: Training and validation loss comparison. Both runs converge stably, and `block_size=32` finishes with lower validation loss.

- token-level NLL and PPL on a fixed 2000-sample test slice,
- final validation `eval_loss`,
- fixed-prompt before/after generations.

Cross-run `train_loss` is not treated as primary comparison evidence because resumed-run behavior affects direct comparability.

This protocol separates optimization signal (NLL/PPL and `eval_loss`) from user-facing behavior (prompt-level quality).

3 Results and Analysis

OFT (main) improves over base by 39.39% in NLL and 72.33% in PPL. Against ablation, `block_size=32` improves by 1.09% in NLL and 2.16% in PPL (relative). The strongest quantitative highlight is the large base-to-OFT gap under fixed compute budget.

Integrated discussion. Gains are task-dependent: concise constrained tasks improve clearly, while reasoning depth and strict format compliance may regress.

Structured analysis (P1/P2/P3).

- **P1: Primary effectiveness signal.** Under fixed compute and split, OFT shows a large in-domain gain over base (NLL 3.2619 → 1.9770; PPL 26.0990 → 7.2207), consistent across table metrics and curve trends.
- **P2: Robustness under constrained ablation.** With matched settings, `block_size=32` stays better than `16` (NLL -1.09%, PPL -2.16%, and lower `eval_loss`); the margin is modest but directionally stable.
- **P3: Behavioral tradeoff and deployment implication.** Concise structured tasks improve (P007/P010/P033), while format-critical or depth-demanding prompts can regress (P016/P042/P045), suggesting extra guardrails for strict-format and reasoning-depth use cases.

4 Limitations and Conclusion

Limitations.

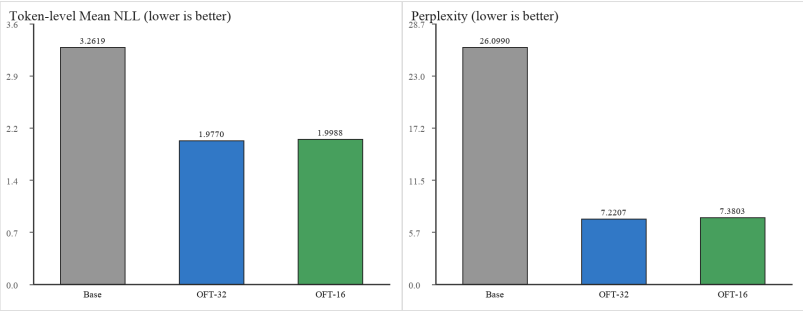

Figure 2: Token-level NLL/PPL bars for base, main, and ablation, showing a large base-to-OFT gap and a smaller but consistent main-vs-ablation gap.

Table 2: Representative before-vs-after qualitative evidence with improvements, regressions, and tradeoff behavior.

Prompt ID	Task	Outcome	Base snippet	OFT snippet
P007	Classification	Improved	Explanatory sentence around sentiment.	Label-only output matching requested format.
P010	Extraction	Improved	Verbose extraction with surrounding explanation.	Compact field lines: location / time / person.
P033	Safety	Improved	Long risk explanation and general advice.	Direct refusal plus concise safe redirection.
P025	Coding	Tradeoff	Correct code with extra explanation/tests.	Correct concise function, less pedagogy.
P016	Reasoning	Regressed	Multi-step operational incident response.	Short checklist bullets with less depth.
P042	Instruction-following	Regressed	Three-line fruit list (format correct).	One-line comma-separated output (format wrong).
P045	Domain explanation	Regressed	Multi-point rationale for canary release value.	Single-sentence compressed summary.

- Evaluation is in-domain; broad transfer claims are unsupported.
- Potential near-duplicate overlap can inflate measured gains.
- NLL/PPL do not fully capture preference quality, factuality, or strict format reliability.
- Qualitative evidence is representative but not exhaustive.

These results should be interpreted as *task- and distribution-scoped* rather than universal model improvement.

Conclusion.

- OFT achieves strong in-domain likelihood improvement over the frozen base model.
- `oft_block_size=32` outperforms `16` across tracked primary metrics.
- The main practical tradeoff is improved conciseness vs. occasional depth/format regressions.

Instruction alignment statement. This report includes all required evidence: training-loss curves (Figure 1), final quantitative performance (Table 1), and before-vs-after qualitative outcomes (Table 2).

References

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